

# Week 6: Integration I

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# Forget calculation rules for now ...

Forget the calculation rules:

$$\int x \, dx = \frac{1}{2}x^2 + C.$$

How do we then determine the area between  $f(x) = x$  and the  $x$ -axis:

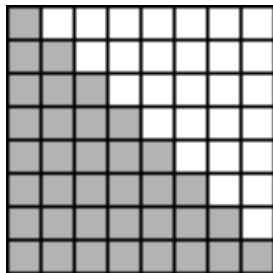
$$\int_0^1 x \, dx = \left[ \frac{1}{2}x^2 \right]_0^1 = 1/2 - 0 = 1/2.$$

without these rules?

Answer: By approximating the area with a lot of rectangles!

# Sum notation $\Sigma$ for $1 + 2 + \dots + n$

$$1 + 2 + \dots + n = ?$$

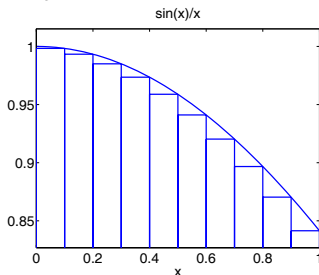


$$1 + 2 + \dots + n = \sum_{k=1}^n k = \frac{1}{2}n \cdot n + \frac{1}{2}n = \frac{1}{2}n(n+1)$$

Ex:  $1 + 2 + \dots + 99 + 100 = \sum_{k=1}^{100} k = \frac{1}{2} \cdot 100 \cdot 101 = 5050$

# Integral notation $\int$ areas

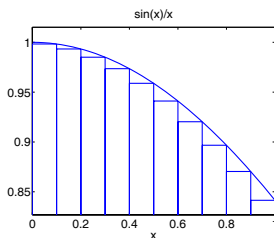
Definite integral:  $\int_a^b f(x) dx = \text{"area between graph and x-axis"}$



Approximate the area with help of eg 10 rectangles  $R_1, R_2, \dots, R_{10}$ :

- 1 Divide  $[0, 1]$  in  $n = 10$  intervals of the same width  $1/n = 0.1$ ;
- 2 Make 10 rectangles that touch the graph with the upper right corner
- 3 Add the areas of the 10 rectangles that touch the graph:

# Approximate integral $\int$ (area) with $\sum$



The areas of the rectangles  $R_1, \dots, R_{10}$  approximate the area below the graph:

$$\begin{aligned} \int_0^1 f(x) dx &\approx \underbrace{0.1 \cdot f(0.1)}_{A(R_1)} + \underbrace{0.1 \cdot f(0.2)}_{A(R_2)} + \cdots + \underbrace{0.1 \cdot f(1)}_{A(R_{10})} \\ &= \underbrace{\sum_{k=1}^{10} 0.1 \cdot f\left(\frac{k}{10}\right)}_{\text{Riemann sum}} \end{aligned}$$

# Definition of integral $\int_a^b f(x) dx$ : A limit

For a function  $f$  on  $[a, b]$  approximate the area between  $f$  and the  $x$ -axis:

- 1 Divide  $[a, b]$  in eg  $n$  intervals of the same width  $h = (b - a)/n$ ;
- 2 Make  $n$  rectangles that touch the graph with the upper right corner
- 3 Add the areas of the  $n$  rectangles that touch the graph:

$$\int_a^b f(x) dx \approx \underbrace{h \cdot f(a+h)}_{A(R_1)} + \underbrace{h \cdot f(a+2h)}_{A(R_1)} + \cdots + h \cdot \underbrace{f(a+n \cdot h)}_{=b} = \sum_{k=1}^n hf(a+k \cdot h)$$

If  $f$  is continuous: Take more and more rectangles  $\implies$  more accurate approximation and

$$\int_a^b f(x) dx \stackrel{\text{definition}}{=} \lim_{n \rightarrow \infty} \sum_{k=1}^n \underbrace{\frac{b-a}{n}}_{\text{width}} \cdot \underbrace{f\left(a + k \cdot \frac{b-a}{n}\right)}_{\text{height}}$$

# Integral calculation with limit

Question: For  $f(x) = x$  determine with the limit definition

$$\begin{aligned}\int_0^1 x \, dx &= \int_0^1 f(x) \, dx \quad \underset{a=0, b=1}{=} \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1-0}{n} \cdot f\left(0 + k \cdot \frac{1-0}{n}\right) \\ &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{n} \cdot f\left(\frac{k}{n}\right) \\ &= \lim_{n \rightarrow \infty} \frac{1}{n} \cdot \sum_{k=1}^n f\left(\frac{k}{n}\right) \\ &\underset{f(x)=x}{=} \lim_{n \rightarrow \infty} \frac{1}{n} \cdot \sum_{k=1}^n \frac{k}{n} \\ &= \lim_{n \rightarrow \infty} \frac{1}{n^2} \cdot \sum_{k=1}^n k\end{aligned}$$

# Integral calculation with limit

Remember: For  $f(x) = x$  determine with the limit definition

$$\begin{aligned}\int_0^1 x \, dx &= \dots = \lim_{n \rightarrow \infty} \frac{1}{n^2} \cdot \sum_{k=1}^n k \\&= \lim_{n \rightarrow \infty} \frac{1}{n^2} \cdot \left( \frac{1}{2} n \cdot (n+1) \right) = \lim_{n \rightarrow \infty} \frac{1}{2} \frac{n \cdot (n+1)}{n^2} \\&= \lim_{n \rightarrow \infty} \frac{1}{2} \frac{n^2 + n}{n^2} = \frac{1}{2} \cdot \lim_{n \rightarrow \infty} \frac{n^2 + n}{n^2} \\&= \frac{1}{2} \cdot \lim_{n \rightarrow \infty} \left( 1 + \frac{1}{n} \right) = \frac{1}{2} \cdot 1 = \frac{1}{2}\end{aligned}$$

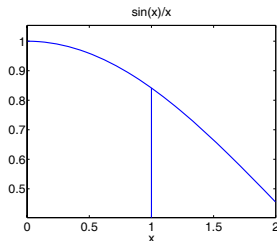
so indeed:  $\int_0^1 x \, dx = \left[ \frac{1}{2} x^2 \right]_0^1 = \frac{1}{2} 1^2 - \frac{1}{2} 0^2 = \frac{1}{2}.$

# Properties of the integral

The limit definition for  $\int_a^b f(x) dx$  leads to

$$\int_a^b f(x) dx + \int_b^c f(x) dx = \int_a^c f(x) dx$$

Ex:  $\int_0^2 \frac{\sin(x)}{x} dx = \int_0^1 \frac{\sin(x)}{x} dx + \int_1^2 \frac{\sin(x)}{x} dx$

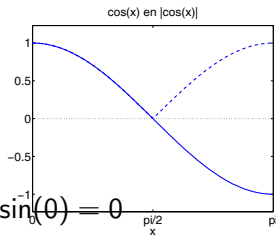


# Properties of the integral

$$\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$$

Ex:  $\underbrace{\left| \int_0^\pi \cos(x) dx \right|}_{=0} \leq \underbrace{\int_0^\pi |\cos(x)| dx}_{=2}$  because

$$\int_0^\pi \cos(x) dx = [\sin(x)]_0^\pi = \sin(\pi) - \sin(0) = 0$$

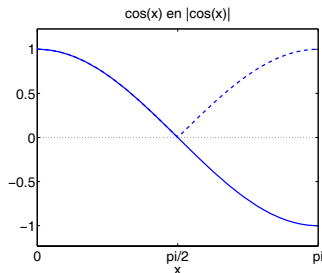


and

$$\begin{aligned} \int_0^\pi |\cos(x)| dx &= \underbrace{\int_0^{\frac{1}{2}\pi} |\cos(x)| dx}_{\cos(x) \geq 0, x \in [0, \pi/2]} + \underbrace{\int_{\frac{1}{2}\pi}^\pi |\cos(x)| dx}_{\cos(x) \leq 0, x \in [\pi/2, \pi]} \\ &= \int_0^{\frac{1}{2}\pi} \cos(x) dx + \int_{\frac{1}{2}\pi}^\pi -\cos(x) dx \\ &= 1 - (-1) = 2 \end{aligned}$$

# Properties of the integral

Ex: Integral with absolute value: Determine where  $f(x) = 0$ :



$$\int_0^{3\pi/2} |\cos(x)| dx = \int_0^{\pi/2} \cos(x) dx + \int_{\pi/2}^{3\pi/2} (-\cos(x)) dx$$

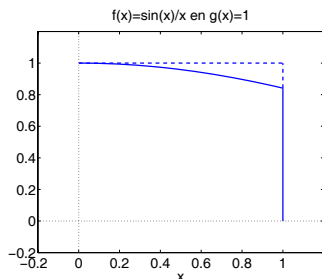
# Properties of the integral

Area **under** 2 graphs:

If  $a \leq b$  and  $f(x) \leq g(x)$  for  $x \in [a, b]$  then

$$\int_a^b f(x) dx \leq \int_a^b g(x) dx$$

Ex: 
$$\int_0^1 \frac{\sin(x)}{x} dx \leq \int_0^1 1 dx = 1$$

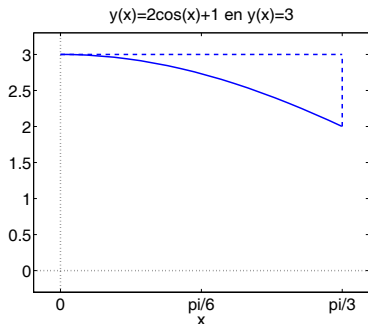


# Properties of the integral

Area **between** 2 graphs: If  $f(x) \geq g(x)$  on  $[a, b]$ ,

then the area between the graphs of  $f$  and  $g$  is:

$$\int_a^b (f(x) - g(x)) dx$$



$$\text{area} = \int_0^{\pi/3} (3 - (2\cos(x) + 1)) dx$$

One is calculating the difference of two areas.

# Properties of the integral

If  $f$  is an odd function

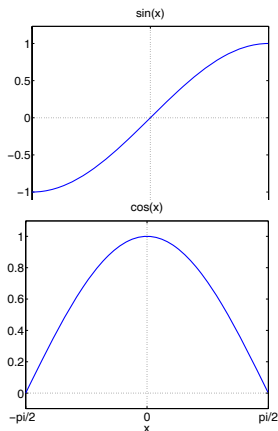
then  $\int_{-a}^a f(x) dx = 0$

Ex:  $\int_{-\pi/2}^{\pi/2} \sin(x) dx = 0$

If  $f$  is an even function

then  $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$

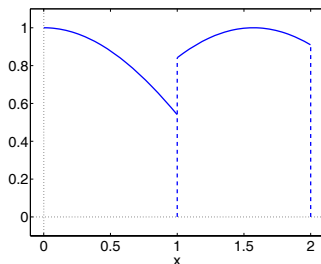
Ex:  $\int_{-\pi/2}^{\pi/2} \cos(x) dx = 2 \int_0^{\pi/2} \cos(x) dx$



# Properties of the integral (VI)

$\int$  for piecewise continuous function: divide in pieces and add:

deelsgewijs continue functie



Ex:  $\int_a^b f(x) dx = \int_a^{c_1} f(x) dx + \int_{c_1}^{c_2} f(x) dx + \int_{c_2}^b f(x) dx$

# Properties of the integral

- ▶  $\int_a^a f(x) dx = 0$
- ▶  $\int_a^b f(x) dx = - \int_b^a f(x) dx$
- ▶  $\int_a^b cf(x) dx = c \cdot \int_a^b f(x) dx$
- ▶  $\int_a^b (f(x) + g(x)) dx = \int_a^b f(x) dx + \int_a^b g(x) dx$
- ▶  $\int_a^b f(x) dx + \int_b^c f(x) dx = \int_a^c f(x) dx$
- ▶ If  $a \leq b$  and  $f(x) \leq g(x)$  for  $x \in [a, b]$  then  $\int_a^b f(x) dx \leq \int_a^b g(x) dx$
- ▶ If  $f$  is an odd function, then  $\int_{-a}^a f(x) dx = 0$
- ▶ If  $f$  is an even function, then  $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$
- ▶  $\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$

# Average value of a series of numbers and functions

Average value of a series of numbers:

$$\frac{1}{n}(a_1 + a_2 + \cdots + a_n) = \frac{1}{n} \sum_{k=1}^n a_k$$

“Integral = infinite sum”

Average value of a function on interval:

$$\frac{1}{b-a} \int_a^b f(x) dx$$

# Primitive functions

Assume  $F(x)$  is a function with

$$F'(x) = f(x)$$

as derivative. Then  $F(x)$  is called a primitive (function) of  $f(x)$

If  $F(x)$  is a primitive of  $f(x)$ , then is  $F(x) + C$  and also:

$$\frac{d}{dx} (F(x) + C) = \underbrace{\frac{d}{dx} F(x)}_{f(x)} + \underbrace{\frac{d}{dx} C}_0 = f(x)$$

so a primitive is only determined up to a constant  $+ C$ !

# Fundamental theorem of calculus

So do not say **the** primitive of  $f$  but **a** primitive of  $f$ . **Ex:**  $f(x) = x$  has as primitive  $\frac{1}{2}x^2 + C$ ,  $C$ : integration constant

Assume  $F$  is a primitive function of  $f$  (thus  $F' = f$ ):

## Fundamental theorem of calculus

$$\int_a^b f(x) dx = F(b) - F(a)$$

# Primitives

| $f$                       | $F = \int f + C$                              |
|---------------------------|-----------------------------------------------|
| $x^r$                     | $\frac{1}{r+1}x^{r+1} + C \quad (r \neq -1)$  |
| $\frac{1}{x}$             | $\ln x  + C$                                  |
| $\sin(ax)$                | $-\frac{1}{a}\cos(ax) + C$                    |
| $\cos(ax)$                | $\frac{1}{a}\sin(ax) + C$                     |
| $a^x$                     | $\frac{1}{\ln(a)}a^x + C$                     |
| $e^{ax}$                  | $\frac{1}{a}e^{ax} + C$                       |
| $\frac{1}{\cos^2(x)}$     | $\tan(x) + C$                                 |
| $\frac{1}{1+x^2}$         | $\arctan(x) + C$                              |
| $\frac{1}{\sqrt{1-x^2}}$  | $\arcsin(x) + C$                              |
| $-\frac{1}{\sqrt{1-x^2}}$ | $\arccos(x) + C$                              |
| $\tan(x)$                 | $-\ln \cos(x)  + C \quad (\text{see[S 5.6]})$ |

# Quiz: Integration and primitives

- ▶  $\int_{-2}^2 \sqrt{4 - x^2} \, dx$
- ▶  $\int_0^{\ln(3)} e^{3x} \, dx$
- ▶  $\int_0^3 (\sqrt{2} + 1)x^{\sqrt{2}} \, dx$
- ▶  $\int_{-3}^{-2} \frac{dx}{x} = \int_{-3}^{-2} \frac{1}{x} \, dx$
- ▶  $\int_{-2}^2 (3x + 2)^2 \, dx$
- ▶  $\int_{-\pi/5}^{\pi/5} (x^2 + \sin(x)) \, dx$
- ▶  $\int_1^8 x^{-5/3} \, dx$
- ▶ average value of  $f(x) = (x - 7)^2$  on  $[0, 12]$ ?

# Substitution method

Consider the example

$$\int \cos(x^2) 2x \, dx$$

We can't see a primitive of  $f(x) = \cos(x^2) 2x$ .

But we see that  $2x$  is the derivative of  $g(x) = x^2$ .

$$\begin{aligned} \int \cos(x^2) 2x \, dx &= \int \cos(g(x)) g'(x) \, dx \stackrel{u=g(x)}{=} \int \cos(u) \, du \\ &= \sin(u) + C = \sin(x^2) + C \end{aligned}$$

Why is  $g'(x) \, dx = du$ ?

$$g'(x) = \frac{d}{dx} g(x) \stackrel{u=g(x)}{=} \frac{d}{dx} u = \frac{du}{dx} \stackrel{\cdot dx}{\Leftrightarrow} g'(x) \, dx = du$$

## Substitution rule

$$\int_a^b f(g(x))g'(x) dx \stackrel{u=g(x)}{=} \int_{g(a)}^{g(b)} f(u) du$$

How to determine  $\int_0^2 f(x) dx = \int_0^2 \frac{x}{\sqrt{1+x^2}} dx$

The function in the integral is not immediately obvious to be of type  $f(g(x))g'(x)$ ... **but it doesn't have to be!**

# Integrals with substitution

To determine  $\int_0^2 f(x) dx = \int_0^2 \frac{x}{\sqrt{1+x^2}} dx = \int_0^2 \frac{x}{\sqrt{1+x^2}} dx$

We can't see a primitive of  $f$  – and try substitution:

$\boxed{x^2 = u} \implies x = \pm\sqrt{u}$  and thus also  $\frac{x}{\sqrt{1+x^2}} = \frac{\pm\sqrt{u}}{\sqrt{1+u}}$  Further

$$\begin{aligned} \frac{d}{dx} u &= \frac{d}{dx} x^2 \implies \frac{du}{dx} = 2x \implies du = 2x dx \implies \\ dx &= \frac{1}{2x} du = \frac{1}{2 \pm\sqrt{u}} du \end{aligned}$$

Also  $x = 0 \implies u = 0^2 = 0$  and  $x = 2 \implies u = 4$

Together

$$\begin{aligned} \int_0^2 \frac{x}{\sqrt{1+x^2}} dx &= \int_0^4 \frac{\pm\sqrt{u}}{\sqrt{1+u}} \frac{1}{2 \pm\sqrt{u}} du = \frac{1}{2} \int_0^4 \frac{1}{\sqrt{1+u}} du \\ &= \frac{1}{2} [2\sqrt{1+u}]_0^4 = \sqrt{5} - \sqrt{1} \end{aligned}$$

## Substitution rule

$$\int_a^b f(g(x))g'(x) dx \stackrel{u=g(x)}{=} \int_{g(a)}^{g(b)} f(u) du$$

Step-by-step guide integration with substitution:

- 1 Replace  $g(x)$  by  $u$  (substitute  $u = g(x)$ )
- 2 Replace  $g'(x) dx$  by  $du$
- 3 See to it that you also replace all remaining  $x$  by  $u$
- 4 Evt: boundaries  $x \rightarrow$  boundaries for  $u$   
if boundaries are given, and this is possible

# Substitution tips

- ▶ Fractions  $\int \frac{t(x)}{n(x)}$ : Try  $u = n(x)$
- ▶ Fractions  $\int \frac{t(x)}{\sqrt{n(x)}}$ : Try  $u = n(x)$  or  $u = \sqrt{n(x)}$
- ▶ Powers  $\int \cos(x)e^{\sin(x)}$ : Try exponent  $u = \sin(x)$
- ▶ Chain rule  $\int h(x) \cdot f'(x) \cdot f(x)$ : Try  $u = f(x)$ . **Ex:** If  $f(x) = \arccos(x)$  then is  $f'(x) = -\frac{1}{\sqrt{1-x^2}}$  in

$$\int -\frac{(\arccos(x))^4}{\sqrt{1-x^2}} dx$$

- ▶ Practice! ... Practice more!

# Primitives with substitution

Difference between calculating primitives and integrals:

- 1 skip integration boundaries
- 2 add integration constant  $+ C$  !
- 3 back substitution

# Primitives with substitution

To determine  $\int \frac{x^5}{2-x^6} dx = \int \frac{1}{2-x^6} x^5 dx$

We can't see a primitive of  $f$  – and try substitution:

$$\boxed{2-x^6 = u} \text{ Further}$$

$$\begin{aligned}\frac{d}{dx} u &= \frac{d}{dx} (2-x^6) \implies \frac{du}{dx} = -6x^5 \implies \\ du &= -6x^5 dx \implies x^5 dx = -\frac{1}{6} du\end{aligned}$$

Integration boundaries: not needed. Together

$$\begin{aligned}\int \frac{x^5}{2-x^6} dx &= \int \frac{1}{2-x^6} x^5 dx = \int \frac{1}{u} \cdot -\frac{1}{6} du \\ &= -\frac{1}{6} \int \frac{1}{u} du \\ &= -\frac{1}{6} \ln |u| + C = -\frac{1}{6} \ln |2-x^6| + C\end{aligned}$$

# Primitives with substitution

To determine  $\int x^3 \sqrt{x^2 + 2} dx \left( = \int x^2 \sqrt{x^2 + 2} \cdot x dx \right)$

We can't see a primitive of  $f$  – and try substitution:

$$\boxed{x^2 + 2 = u} \text{ Further}$$

$$\begin{aligned} \frac{d}{dx} u &= \frac{d}{dx} (x^2 + 2) \implies \frac{du}{dx} = 2x \implies \\ du &= 2x dx \implies x dx = \frac{1}{2} du \end{aligned}$$

Integration boundaries: not needed. Together

$$\begin{aligned} \int x^3 \sqrt{x^2 + 2} dx &= \int x^2 \sqrt{x^2 + 2} x dx = \int (u - 2) \sqrt{u} \cdot \frac{1}{2} du \\ &= \frac{1}{2} \int u^{3/2} - 2u^{1/2} du \\ &= \frac{1}{2} \left( \frac{2}{5} u^{5/2} - \frac{4}{3} u^{3/2} \right) + C \\ &= \frac{1}{2} \left( \frac{2}{5} (x^2 + 2)^{5/2} - \frac{4}{3} (x^2 + 2)^{3/2} \right) + C \end{aligned}$$

# Quiz: The substitution rule

- ▶  $\int \frac{x^5}{\sqrt{2-x^6}} dx$
- ▶  $\int x \sqrt[5]{49+x^2} dx$
- ▶  $\int \sin(3x-2) dx$  “direct” and with substitution
- ▶  $\int x^2 \left( \frac{x^3}{27} + 3 \right)^8 dx$
- ▶  $\int \frac{3}{x^4} \cos \left( \frac{1}{x^3} - 2 \right) dx$
- ▶  $\int \frac{1}{(x+3)\ln(x+3)} dx$
- ▶  $\int \frac{21}{4+25x^2} dx$
- ▶  $\int_0^{3\sqrt{3}} \frac{13}{\sqrt{36-x^2}} dx$
- ▶  $\int_0^{\pi/4} (1 - \cos(2x)) \sin(2x) dx$

# Integration/primitives with cos and sin

To determine  $\int f(x) dx = \int \frac{6 \sin(6x)}{5 - \cos(6x)} dx$

We can't see a primitive of  $f$  – and try substitution:

$$\boxed{5 - \cos(6x) = u} \text{ Further}$$

$$\begin{aligned} \frac{d}{dx} u &= \frac{d}{dx} (5 - \cos(6x)) \implies \frac{du}{dx} = 6 \sin(6x) \implies \\ \text{green } du &= \text{pink } 6 \sin(6x) dx \end{aligned}$$

Integration boundaries: don't need. Together

$$\begin{aligned} \int \frac{6 \sin(6x)}{5 - \cos(6x)} dx &= \int \frac{1}{5 - \cos(6x)} \text{pink } 6 \sin(6x) dx = \int \frac{\text{brown } 1}{\text{brown } u} \cdot \text{green } du \\ &= \ln |u| + \text{pink } C = \ln |5 - \cos(6x)| + \text{pink } C \end{aligned}$$

$$\int \tan(x) dx = \int \frac{\sin(x)}{\cos(x)} dx$$

Substitution:  $u = \cos(x)$ , then  $du = -\sin(x)dx$ , thus:

$$\int \frac{\sin(x)}{\cos(x)} dx = \int \frac{-1}{u} du = -\ln |u| = -\ln |\cos(x)|$$

$$\begin{aligned} \int \tan^2(x) dx &= \int \frac{\sin^2(x)}{\cos^2(x)} dx = \int \frac{1-\cos^2(x)}{\cos^2(x)} dx \\ &= \int \left( \frac{1}{\cos^2(x)} - 1 \right) dx = \tan(x) - x \end{aligned}$$

# Integrals with $\sin^m(x)$ , $\cos^n(x)$

$$\int \sin^4(x) \cos(x) dx = \frac{1}{5} \sin^5(x)$$

Too fast?  $u = \sin(x)$  then  $du = \cos(x) dx$  and

$$\int \sin^4(x) \cos(x) dx = \int u^4 du = \frac{1}{5} u^5 = \frac{1}{5} \sin^5(x)$$

$$\begin{aligned} \int \sin^4(x) \cos^3(x) dx &= \int \sin^4(x) (1 - \sin^2(x)) \cos(x) dx \\ &= \int (\sin^4(x) - \sin^6(x)) \cos(x) dx = \frac{1}{5} \sin^5(x) - \frac{1}{7} \sin^7(x) \end{aligned}$$

With these techniques, all cases with  $\sin^m(x) \cos^n(x)$  are covered if  $m$  or  $n$  (or both) is odd

$$\int \sin^2(x) dx = \int \left( \frac{1}{2} - \frac{1}{2} \cos(2x) \right) dx = \frac{1}{2} x - \frac{1}{4} \sin(2x)$$

$$\int \cos^2(x) dx = \int \left( \frac{1}{2} + \frac{1}{2} \cos(2x) \right) dx = \frac{1}{2} x + \frac{1}{4} \sin(2x)$$

$$\int \sin^4(x) dx = \int (\sin^2(x))^2 dx = \int \left( \frac{1}{2} - \frac{1}{2} \cos(2x) \right)^2 dx$$

$$= \int \left( \frac{1}{4} - \frac{1}{2} \cos(2x) + \frac{1}{4} \cos^2(2x) \right) dx$$

$$\text{and use } \cos^2(2x) = \frac{1}{2} + \frac{1}{2} \cos(4x)$$

# Integrals in $\sin(x)$ and $\cos(x)$

The substitution  $\tan(\theta/2) = x$  transforms  $\sin$  and  $\cos$  to polynomials in  $x$ :

$$\begin{aligned}\cos \theta &= \frac{1 - x^2}{1 + x^2} \\ \sin \theta &= \frac{2x}{1 + x^2} \\ d\theta &= \frac{2}{1 + x^2} dx\end{aligned}$$

# Quiz: Integration

- ▶  $\int_0^{\pi/6} 2^{\cos(3x)} \sin(3x) dx$
- ▶  $\int \frac{1}{\sqrt{23x}(\sqrt{23x+3})} dx$
- ▶  $\int \frac{169}{25+169x^2} dx$
- ▶  $\int \frac{1}{64e^{-6x}+e^{6x}} dx$
- ▶  $\int \frac{1}{\sqrt{1-36x^2}} dx$
- ▶  $\int \frac{1}{\sqrt{36-x^2}} dx$
- ▶  $\int 8(13^{8x+9}) dx$
- ▶  $\int \frac{4x^7}{x^4-5} dx = \int \frac{4x^3 \cdot x^4}{x^4-5} dx$
- ▶  $\int \frac{5-50x}{\sqrt{81-25x^2}} dx$
- ▶  $\int \cos^3\left(\frac{x}{2}\right) dx$
- ▶  $\int_0^{\pi/4} 3x \sqrt{1-\cos(2x)} dx$

# Partial integration

$$\int_a^b f G \, dx = [FG]_a^b - \int_a^b F g \, dx$$

with  $F$  primitive of  $f$  and  $G$  primitive of  $g$  (integration by parts)  
Why?

$$\begin{aligned}(FG)' &= Fg + fG \\ \int_a^b (FG)' \, dx &= \int_a^b Fg \, dx + \int_a^b fG \, dx \\ [FG]_a^b &= \int_a^b Fg \, dx + \int_a^b fG \, dx\end{aligned}$$

$$\int_a^b f G dx = [FG]_a^b - \int_a^b F g dx$$

What to chose for  $f$  and  $G$ ?

- ▶ Rule of thumb: chose as  $G$  the function you want to **simplify**
- ▶ Experience
- ▶ Trial: if it does not work, try a different choice for  $f$  and  $G$
- ▶ Chose  $f = 1$  is there is only one factor  $G$

Of course, the book uses a different notation that can be confused with substitution

# Partial integration

So with or without integration boundaries:

$$\begin{aligned}\int_a^b f G dx &= [FG]_a^b - \int_a^b F g dx \\ \int f G dx &= FG - \int F g dx\end{aligned}$$

Ex:

$$\begin{aligned}\int_0^1 x e^x dx &= \int_0^1 e^x x dx \\ &= \int_0^1 \underbrace{x}_G \underbrace{e^x}_f dx = \left[ \underbrace{e^x}_F \underbrace{x}_G \right]_0^1 - \int_0^1 \underbrace{e^x}_F \underbrace{1}_g dx \\ &= (e - 0) - [e^x]_0^1 = e - (e - 1) = 1\end{aligned}$$

# Partial integration with $f = 1$

Ex: Not a product of 2 functions!

$$\begin{aligned}\int \ln(x) dx &= \int \underbrace{1}_f \underbrace{\ln(x)}_G dx = \underbrace{x}_F \underbrace{\ln(x)}_G - \int \underbrace{x}_F \underbrace{\frac{1}{x}}_g dx \\ &= x \ln(x) - x\end{aligned}$$

The same way also to get: (“p.i.” is partial integration)



$$\begin{aligned}\int \arcsin(x) dx &= \int 1 \cdot \arcsin(x) dx \\ &\stackrel{\text{p.i.}}{=} x \cdot \arcsin(x) - \int x \cdot \frac{1}{\sqrt{1-x^2}} dx \\ &\stackrel{u=1-x^2}{=} x \arcsin(x) + \frac{1}{2} \int \frac{1}{\sqrt{u}} du = \dots\end{aligned}$$

►  $\int \arccos(x) dx$  and  $\int \arctan(x) dx$

## Partial integration (P.I.): Repeat

$$\begin{aligned}\int x^2 e^x dx &= x^2 e^x - \int 2x e^x dx \\&= x^2 e^x - \left( 2x e^x - \int 2e^x dx \right) \\&= x^2 e^x - (2x e^x - 2e^x) \\&= (x^2 - 2x + 2)e^x\end{aligned}$$

## P.I.: Repeating restores integral!

Sometimes you will get back to the start after 2 times partial integration, but this can be useful:

$$\begin{aligned}\int e^{ax} \sin(x) dx &= e^{ax}(-\cos(x)) - \int ae^{ax}(-\cos(x)) dx \\ &\stackrel{\text{p.i.}}{=} -e^{ax} \cos(x) + a \int e^{ax} \cos(x) dx \\ &\stackrel{\text{p.i.}}{=} -e^{ax} \cos(x) + a \left( e^{ax} \sin(x) - \int ae^{ax} \sin(x) dx \right)\end{aligned}$$

Thus  $\int e^{ax} \sin(x) dx = -e^{ax} \cos(x) + ae^{ax} \sin(x) - a^2 \int e^{ax} \sin(x) dx$

and  $(1 + a^2) \int e^{ax} \sin(x) dx = -e^{ax} \cos(x) + ae^{ax} \sin(x)$

so that  $\int e^{ax} \sin(x) dx = \frac{1}{1 + a^2} e^{ax} (-\cos(x) + a \sin(x))$

# Quiz: Partial Integration

- ▶  $\int x \sin(5x) dx$
- ▶  $\int_{1/7}^5 16x \ln(7x) dx$
- ▶  $\int (6x^2 - 5x)e^{5x} dx$
- ▶  $\int_0^{2\pi} x^2 \sin\left(\frac{x}{4}\right) dx$
- ▶  $\int \sin(\ln(5x)) dx = \int 1 \cdot \sin(\ln(5x)) dx$
- ▶  $\int e^{3x} \cos(4x) dx$
- ▶  $\int x \tan^2(x) dx$